AI Vision Monitoring Platform: Use Case Report

Oct 22, 2026 · @Anil Kumar

The platform turns ordinary CCTV into a system that recognizes people, counts crowds, detects hazards and acts on what it sees. Retail and hospitals are the two industries this report recommends first.

Summary

We recommend retail and hospitals as the first two industries, because most of the needed detection already exists in the platform and both have clear, costly problems.

	Retail and malls	Hospitals
Core problem	Theft, unknown footfall, long queues	Patient falls, unauthorized entry to restricted wards, crowding
What we deliver	Footfall and heatmap reports, queue alerts, loss-prevention flags	Fall alerts, ward access control, crowd and wandering alerts
Already built	Person detection, restricted zones, crowd limits, fire and smoke	Restricted zones, face-based staff access, crowd limits, fire, fighting
Still to build	Footfall counter, heatmap, queue length	Fall detection, wandering alert
Best first pilot	One small store with a single entrance	One ward or ICU entrance in a clinic or nursing home

AI agents and automation turn alerts into action: they message the right person, attach the clip, escalate if nobody responds, and write the daily report.

Platform capabilities today

The platform watches one live camera with computer vision, and previews recorder channels without running models. It runs on one PC with a React admin console and a single FastAPI service.

Capability	Status
Person and object detection	Built
Face recognition, staff entry and exit logs	Built
Unknown visitor capture	Built
Restricted areas and overcrowding alerts	Built
Fire, smoke and fighting detection	Built
Unattended bag detection	Built
Emotion and behavior logging (sitting, standing, walking, idle)	Built
Role-based access to every screen	Built
Footfall counting, heatmaps, queue length	To build
Fall detection and wandering alerts	To build
AI agents, messaging and automation	To build
Multi-camera AI and a database	To build

Use case 1: Retail and malls

Retailers lose sales to theft, long queues and blind spots in customer flow, and they usually learn about it only from monthly stock audits. The platform turns store cameras into loss-prevention and analytics tools.

Problems

Problem	Impact
Shoplifting and staff theft	Stock loss found late, at audit time
Unknown footfall	No way to measure conversion from visitors to buyers
Long billing queues	Customers leave without buying
Crowds at weekends and sales	Safety risk and slow response to fire or fights
Uneven staffing	Too few staff at peak hours, too many at quiet ones

Problem	Impact
Entry to godown or after-hours areas	Theft and unauthorized access

Solution and use cases

Use case	What the system does	Status
Footfall counting	Counts people crossing the entrance line, with hourly and daily reports	To build
Heatmap	Shows which sections customers visit and linger in	To build
Queue monitoring	Counts people at a billing counter and alerts when the line passes a set length	Extends the existing area count
Restricted area control	Alerts when anyone not on the allowed list enters the godown or staff area	Built
Overcrowding alert	Alerts when a zone passes its maximum person count	Built
Fire and smoke alert	Detects flames and smoke early	Built
Fight detection	Flags aggressive behavior near entrances and in crowds	Built
Suspicious behavior flag	Flags long loitering or entry into restricted zones for a guard to review	To build

Suspicious behavior is a flag for a human to review, not a verdict that someone is a thief.

Agents and automation

- **Queue agent:** messages the counter manager when a line is too long, for example to open another counter.
- **Footfall report agent:** emails or messages a daily summary with total footfall, peak hours and the most and least visited sections.
- **Loss-prevention agent:** bundles the clip, time, camera and zone into one incident for the guard, who makes the final call.
- **Staffing agent:** suggests extra staff for hours that were busy in previous weeks.
- **Escalation automation:** calls the next manager if an alert is not acknowledged within a set time.

Business value

- Fewer losses and less reliance on guards watching screens.
- Conversion rate from footfall against billing.
- Fewer customers abandoning long queues.
- Staff placed where and when they are needed.

Shopper faces should not be recognized. Use anonymous counting and heatmaps for customers, and keep face matching for staff only.

Use case 2: Hospitals

Hospitals have large areas and limited staff, so falls, unauthorized entry and crowding are noticed late. The platform adds patient safety and access control on top of existing cameras.

Problems

Problem	Impact
Patient falls	Help arrives late, injuries get worse
Unauthorized entry to ICU, OT and NICU	Infection risk, privacy breach, safety risk
Crowds in OPD and emergency	Pushing, fights, slower care
Patients leaving the ward unnoticed	Wandering patients, especially at night
Violence against staff in emergency	Staff safety and disrupted care
Fire and smoke	Patients cannot evacuate quickly
Baby safety in maternity and NICU	Risk of a baby being taken
Manual access and attendance registers	No reliable record of who entered which ward

Solution and use cases

Use case	What the system does	Status
Patient fall alert	Detects a person falling and not getting up, and alerts the ward nurse	To build
ICU, OT and NICU access control	Allows only registered staff and alerts on anyone else, with an entry log	Built

Use case	What the system does	Status
Wandering alert	Alerts when a patient reaches a ward exit zone at night or outside allowed times	To build
OPD and emergency crowding	Counts people in waiting areas and alerts above a limit	Built
Maternity and NICU exit control	Alerts security on unauthorized movement at the exit	Built, using restricted zones
Staff safety	Flags fights and aggressive behavior in emergency	Built
Fire and smoke alert	Detects fire early in pharmacy, kitchen, oxygen store and electrical rooms	Built
Staff attendance by ward	Daily record of who entered which ward and when	Built
Night monitoring	Alerts on unusual movement in corridors, pharmacy and stores	Built, using restricted zones

Agents and automation

- **Fall response agent:** sends the nurse a clip with ward and bed number, and escalates to the charge nurse if nobody acknowledges within a set time.
- **Access agent:** keeps the entry record for restricted wards and alerts security on unauthorized entry.
- **Wandering agent:** informs the duty nurse and security when a patient moves toward an exit.
- **Crowd agent:** tells reception and security when a waiting area is overcrowded.
- **Incident report agent:** records every alert with time, ward and clip for audits.
- **Daily report agent:** sends management a summary of incidents, attendance and crowding.

Safeguards

- Place cameras in corridors, entrances and common areas. Never inside patient rooms or washrooms.
- The system only raises alerts. Doctors and nurses make every medical decision.
- Collect staff face data with consent, and do not store patient faces unless required.

- Expect false alarms from fall detection at first, and tune thresholds with nurses during the pilot.
- The platform runs on a local PC, so video can stay inside the hospital network.

AI agents and automation

Today the platform sees and alerts. Agents let it act, so that every alert reaches the right person with the evidence attached.

Capability	What it does	Retail	Hospitals
Auto-response	Sends WhatsApp, SMS or a call with the clip and location	Guard, counter manager	Nurse, security
Escalation	Calls the next person if an alert is not acknowledged in time	Store manager	Charge nurse
Incident writer	Builds a record with time, camera, zone and clip	Loss-prevention cases	Audit records
Natural-language search	Answers questions like who entered a room last night	Godown access	ICU access
Daily reports	Emails a summary of the day	Footfall, peak hours	Incidents, attendance
Pattern learning	Learns normal activity and flags unusual movement	After-hours movement	Night corridor movement

Platform upgrades needed to scale

The current design suits a single camera on one PC. Both industries need the following before multi-camera rollouts.

Today	Upgrade
One AI camera at a time	A worker per camera, or a GPU server
JSON files for storage	A database such as PostgreSQL for events, footfall and audit records
Browser polling	WebSocket or an event queue for live alerts
No person tracking	A tracker, needed for footfall, heatmaps and fall detection

Today	Upgrade
No outbound messaging	WhatsApp, SMS or call integration
One PC	Separate API and vision servers, with restart and backup for 24x7 use

Roadmap, risks and compliance

Start with a small pilot where the current platform already works, then add features in phases.

Phase	Retail	Hospitals
1. Pilot	Footfall counter, queue alert, restricted area, fire and smoke in one small store	Access control, fire and smoke, crowd alert at one ward or ICU entrance
2. Expand	Heatmap, daily report agent, escalation	Fall detection and wandering alert, tuned with nurses
3. Scale	Suspicious behavior flag, search agent, multi-store dashboard	Multi-ward dashboard, search agent, daily reports

Risks and compliance

- **Privacy law:** India's DPDP Act applies to face and video data. Display camera notices and keep a privacy policy.
- **Consent:** collect staff face data with consent, and do not recognize customers or patients unless it is necessary.
- **False alarms:** fall detection and suspicious behavior flags will produce false alerts at first, so tune them in the pilot.
- **Human decision:** the system flags events and people decide what to do. It does not accuse or diagnose.
- **Pilot data:** value figures such as loss reduction or queue improvement should come from pilot measurements, not estimates.